

Global Intelligence Summary

Information Epidemiology

Tracking the Digital Contagion of
Maternal Vaccine Misinformation

Synthesised from a 6-month, 15-country social media surveillance operation.



“Vaccine hesitancy is one of the ten global health threats.”

(World Health Organization, 2019)



The Target: Maternal vaccines (influenza, pertussis, tetanus).

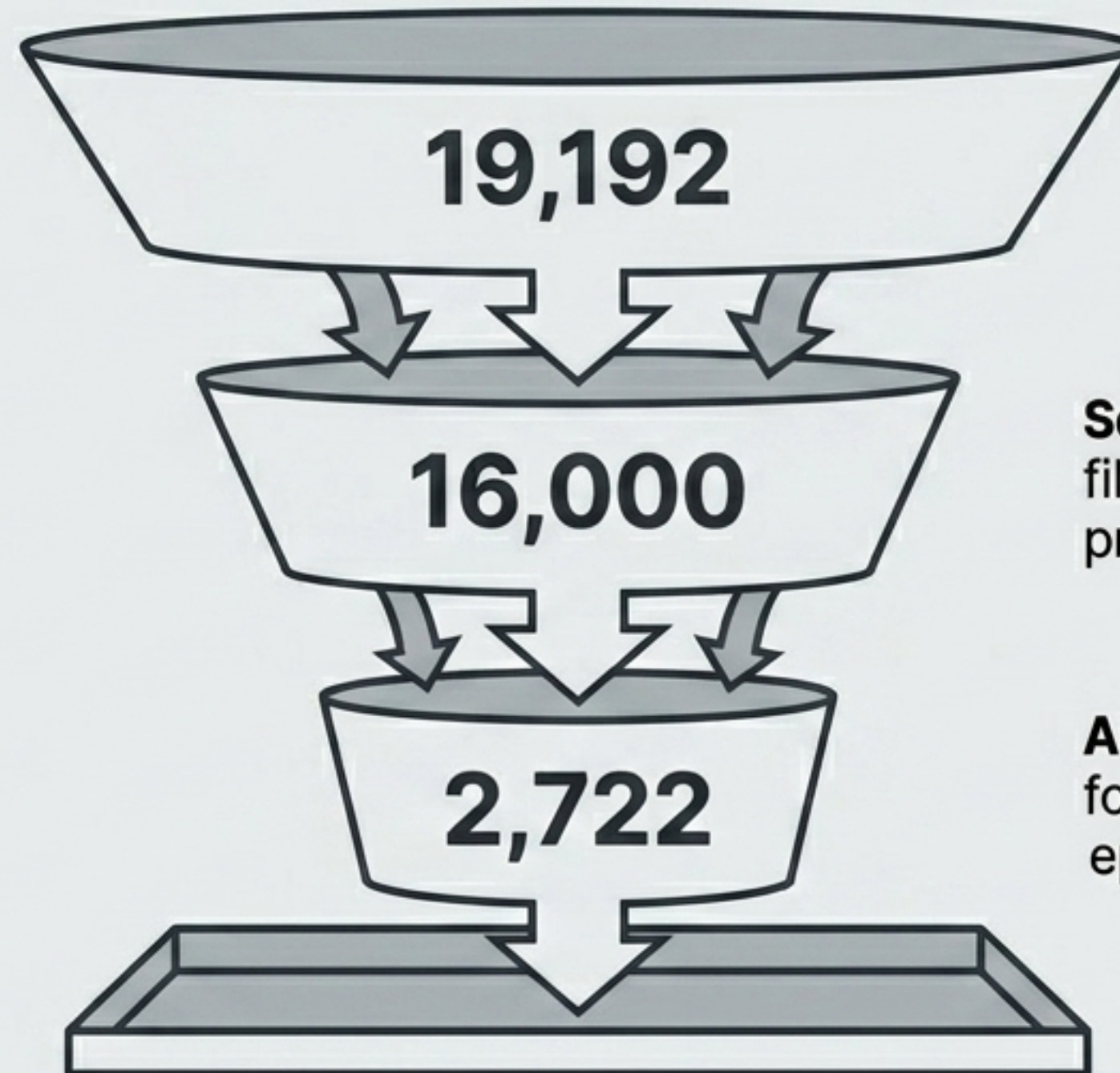


The Risk: Severe morbidity and mortality for neonates and mothers.



The Reality: Despite universal WHO recommendations, global uptake remains dangerously inadequate due to viral digital misinformation.

Global Surveillance: The Distillation Process



Raw Signals. Social media posts extracted via Meltwater – across 15 countries and 16 languages (Nov 2018 – Apr 2019).

Screened Posts. Deduplicated and filtered for specific focus on pregnancy-administered vaccines.

Annotated Nodes. Manually coded for stance by three independent epidemiologists with a 0.56 Fleiss agreement score.

The Stance Diagnostic Matrix



Promotional (Vibrant Green):

Encourages maternal vaccination; highlights benefits.

Real Quote: "It's flu season. Getting the flu jab while pregnant is the best way to protect mum and baby."



Neutral (Cool Grey):

Factual recommendations; no uncertainty or emotion.

Real Quote: "Tdap vaccines during pregnancy: Find out more: @link"



Ambiguous (Amber):

Indecision; seeking guidance; mixed mixed information.

Real Quote: "I got a flu jab... but I just read some tweets saying vaccines cause autism(!) Is this true?"

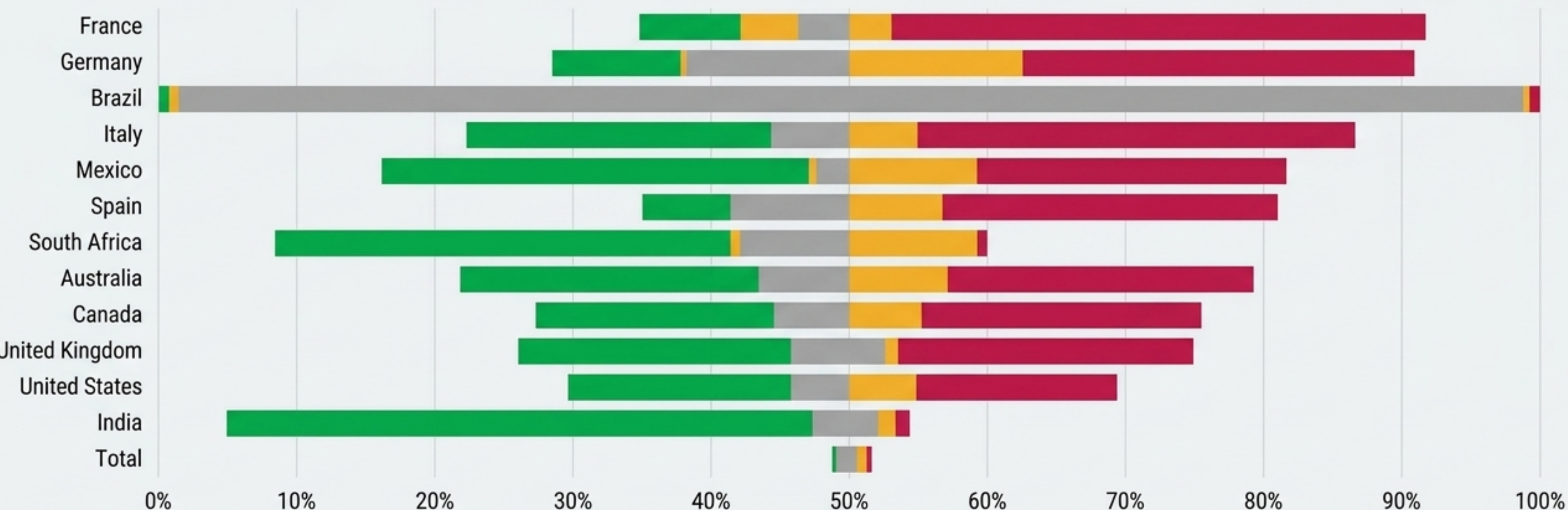


Discouraging (Crimson):

Negative attitudes; questions safety; discourages uptake.

Real Quote: "Vaccines for pregnant women...?! Absurd"

Mapping the Global Immune System



Callout Left (High Promotional):

India (88% Green), South Africa (85% Green).

Driven by strong public health messaging.

Callout Center (Neutral/Ambiguous):

Brazil (99% Grey), Germany (53% Grey, 27% Amber).

Information-seeking environments.

Callout Right (High Discouraging):

France (66% Crimson), Italy (45% Crimson), USA (31% Crimson).

High presence of institutional mistrust and alleged safety concerns.

Geographic Discourse Extremes

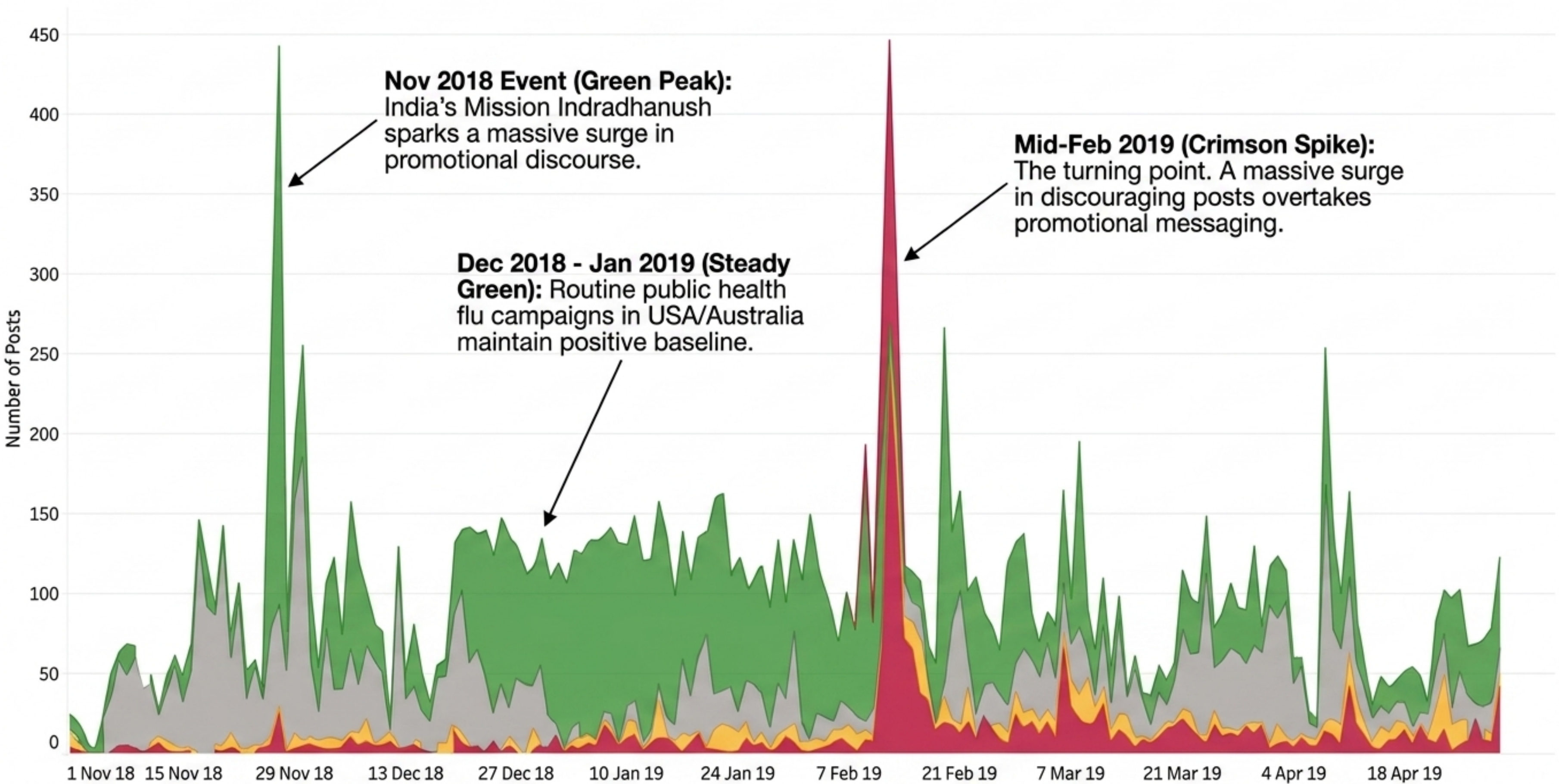
The Sceptics (USA & Italy)

- **Primary Driver:** Mistrust in institutions (CDC, FDA, Italy's ASL).
- **Core Theme:** Extrapolating harm from contraindicated vaccines to recommended ones.
- **Vector Activity:** High presence of bot-associated hashtags.

The Advocates (India & UK)

- **Primary Driver:** Proactive, visible public health campaigns.
- **Core Theme:** Child protection and successful programme launches (e.g., India's Mission Indradhanush).
- **Vector Activity:** Organic tagging of trusted bodies (Public Health England).

The Disinformation Contagion Curve



The Catalyst: February 2019



The Spark: An external NGO (Children's Health Defense) publishes a claim regarding a lawsuit against the US FDA.

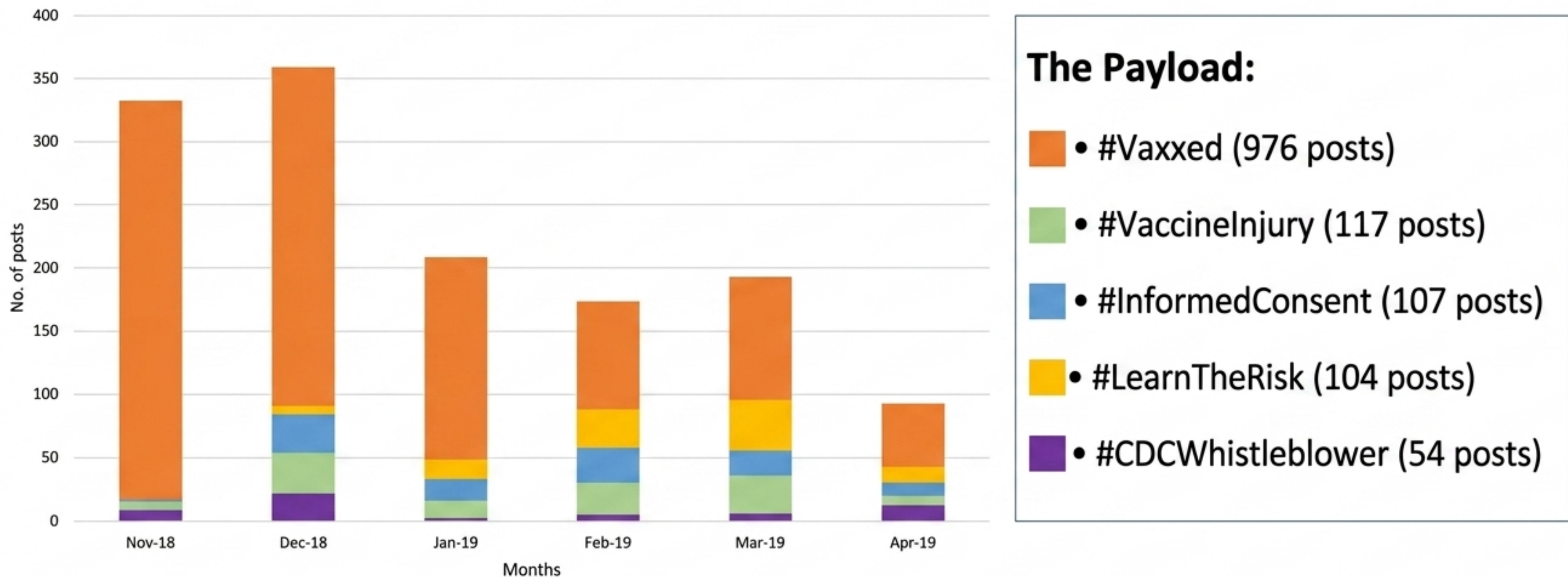
The Narrative: Accuses the FDA of recommending "untested, unlicensed vaccines" for pregnant women.

The Contagion: The rumour immediately crosses borders, shared across 13 of the 15 monitored countries.

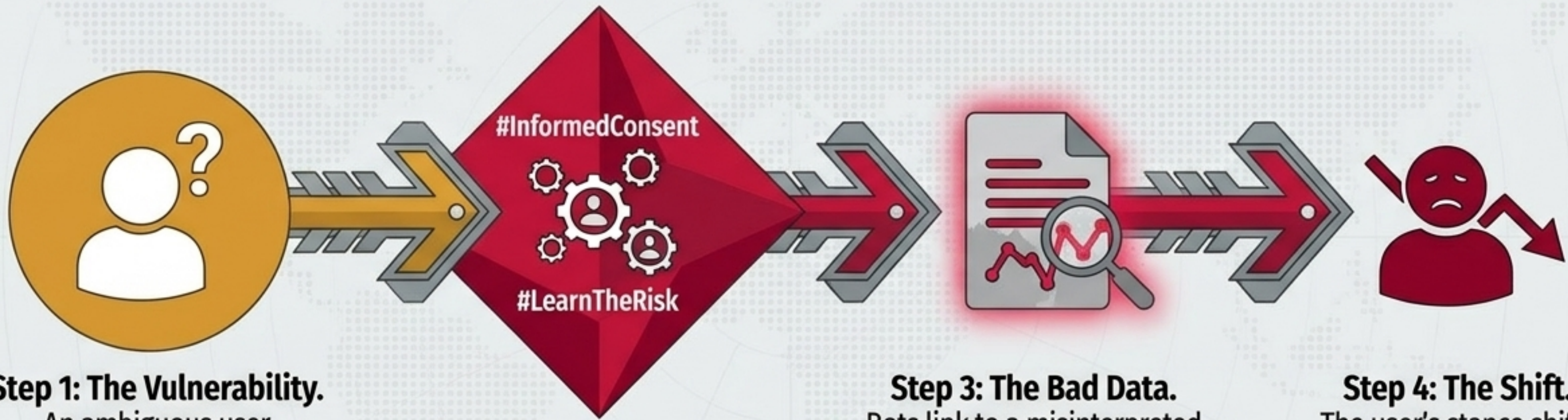
The Impact: Stance in moderate nations shifted dramatically. South Africa's discouraging posts jumped from 7% to 28%; Germany from 5% to 15%.

The Vectors: Weaponised Amplification

Digital outbreaks are rarely entirely organic. Known Russian troll and bot-associated hashtags were deployed to artificially amplify polarisation during vulnerabilities.



The Mechanism of Infection



Step 1: The Vulnerability.

An ambiguous user expresses genuine uncertainty regarding consent and safety data for the flu shot.

Step 2: The Intercept.

Bot accounts hijack the conversation using the #InformedConsent or #LearnTheRisk hashtags.

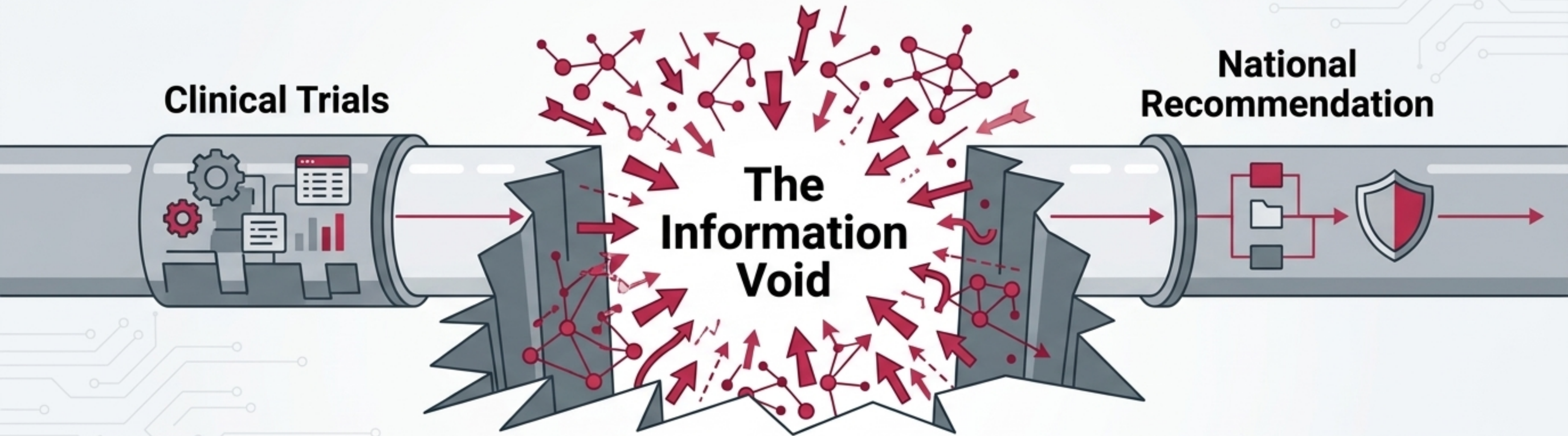
Step 3: The Bad Data.

Bots link to a misinterpreted 2004 study on influenza hospitalisations, stripping it of context to claim the vaccine is entirely ineffective.

Step 4: The Shift.

The user's stance shifts from Ambiguous to Discouraging.

The Root Cause: The Information Void



The Reality:

Due to historical ethical restrictions, pregnant women are rarely included in initial clinical vaccine trials. Recommendations rely heavily on *post-licensure data* and inadvertent vaccination tracking.

The Void:

The public does not understand how post-licensure safety data is generated. Public health bodies have failed to provide accessible, lay-friendly explanations of this process.

The Exploitation:

Misinformation agents exploit this void, accurately claiming vaccines are "unlicensed/off-label" for pregnancy, but maliciously twisting this to mean "untested and unsafe."

The Crisis of Institutional Trust



Distrusted & Denounced:

- **US CDC & FDA:** Accused of hiding safety efficacy data; targeted by organised legal rumours.
- **Italy's ASL:** Regarded as untrustworthy following viral news of contraindicated MMR administration.



Trusted & Tagged:

- **Public Health England (UK) & DAKJ (Germany):** Consistently tagged in promotional statements; viewed as transparent sources of protection advice.

The Antidote: Strategic Recommendations



1. Translate Post-Licensure Data:

Close the "Information Void."
Develop robust, lay-friendly public communications explaining exactly how maternal safety data is researched and verified after licensure.



2. Uniform Consent Guidelines:

Equip healthcare professionals with standardised, transparent verbal and written consent information that proactively addresses historical rumors (e.g., Thalidomide) and real risks.



3. AI-Driven Real-Time Monitoring:

Implement machine-learning dashboards to detect localized discouraging "spikes" in real-time.

A New Paradigm for Public Health

Passive surveillance is dead.

When misinformation is borderless and algorithmically amplified, public health agencies cannot wait for annual surveys. We must meet digital outbreaks with real-time, targeted surges of transparent, mitigating intelligence.

The digital immune system must be as agile as the pathogens it fights.
